

Candidate Number

Candidate Name _____

INTERNATIONAL ENGLISH LANGUAGE TESTING SYSTEM

Academic Reading

Additional materials:

Answer sheet for Listening and Reading

Time 1 hour

INSTRUCTIONS TO CANDIDATES

Do not open this question paper until you are told to do so.

Write your name and candidate number in the spaces at the top of this page.

Read the instructions for each part of the paper carefully.

Answer all the questions.

Write your answers on the answer sheet. Use a pencil.

You must complete the answer sheet within the time limit.

At the end of the test, hand in both this question paper and your answer sheet.

INFORMATION FOR CANDIDATES

There are **40** questions on this question paper.

Each question carries one mark.

READING PASSAGE 1

You should spend about 20 minutes on Questions 1–13, which are based on Reading Passage 1 below.

The development of the bicycle

In a 2005 survey of British radio listeners, the bicycle was voted the most significant technological invention since 1800. As with many other technological innovations, however, it is difficult to determine exactly by whom or when the bicycle was invented. In fact, what we recognise as a bicycle today was never created fully formed, but rather represents a process of evolution, the product of incremental changes initiated by many individuals over time.

The Italian artist and inventor Leonardo da Vinci (1452-1519) is often cited as the first person to visualise the concept of a bicycle, but he never constructed one, and there are doubts surrounding the authenticity of a sketch of a two-wheeled device discovered among his papers. In about 1790, the French Comte de Sivrac was the first person to build what could be said to resemble a bicycle. This was a wooden-framed device known as a Celerifere. With no pedals or steering mechanism, the Celerifere was considered a child's toy rather than a method of adult transport.

In 1817 Baron von Drais, a German inventor, built a device with two wheels of the same size mounted in line on a wooden frame. The users sat on a seat on a frame with a leg on either side of the frame and simply pushed themselves along with their feet. It was known as a Draisienne or hobby horse, and although it could be steered it proved impractical for transportation anywhere but on a smooth road or path.

The next appearance of a two-wheeled device was in 1865, when pedals were attached directly to the front wheel. This device was officially called a velocipede, but was popularly known as the bone shaker as its wooden (later metal) wheels made for an extremely uncomfortable ride. However, riding velocipedes for pleasure became popular, and indoor riding tracks became fairly common in large cities as a result.

The desire for speed was the driving factor behind the next significant step in the development of two-wheeled transport, which took place in the 1870s. The most obvious way to increase the speed that could be obtained was to increase the size of the wheel to which the pedals were attached. The front wheels became larger and larger as manufacturers realised that the bigger the wheel, the further the rider could travel with one rotation of the pedals. At about the same time, the first all-metal two-wheeled devices appeared due to advances in the production of strong, relatively light metal alloys.

Then solid rubber tyres were added, providing a somewhat smoother ride. These newly adapted devices, called High Wheelers, or Penny Farthings, were the first to actually be referred to as bicycles. The rider sat on a seat above the large front wheel, which made them awkward to mount. They were notoriously difficult to ride as they were extremely unstable except when ridden at speed.

Because of the problems riders encountered with high wheelers, an alternative device was specifically designed for females. This version had smaller wheels and incorporated a third wheel for greater stability one in the front and two in the rear. This tricycle was considered more practical for women, since their long dresses were potentially hazardous when riding High Wheelers.

The next important change in bicycle design led to the creation in the 1880s of the basic form of the modern bicycle. The safety bicycle, as the new device was known, saw a return to two same-size wheels. It was powered by pedals in the middle of the frame. These pedals were attached with a chain and gears to the rear wheel. This arrangement meant that the rider, seated on the frame between the wheels, could attain the same speed as with the high wheeler, but without the dangers associated with the huge front wheel. In addition to these design changes, there was the introduction of the pneumatic tyre, developed in 1888 by Scottish veterinary surgeon John Dunlop. This involved a tube inflated with air being inserted inside the hard rubber outer tyre, thus providing a considerably more pleasant ride.

Manufacturers experimented with various designs, but the Rover, created by English inventor and industrialist John Starley, became the most successful and most copied. Virtually all the Rover's features are present on today's bicycles: a diamond-shaped frame, pedals below the saddle that power the back wheel through a chain and gears, and handlebars attached to the front wheel.

The bicycle became incredibly popular in Europe and America in the 1890s, which is often referred to as the Golden Age of Cycling. The bicycle was a practical and relatively inexpensive investment for the working man. It was an efficient means of transport and gave him much greater flexibility in his leisure time. Women, once forced by practical considerations to ride heavy tricycles, could now ride a much more versatile machine that increased their mobility considerably.

Over the past 30 years or so, the world has arguably seen the greatest changes to the bicycle since the safety bicycle was introduced over 100 years ago. The availability of a wider range of materials, improved manufacturing techniques, better scientific knowledge and ever-increasing popularity have led to the introduction of many specialised bicycle models - from commuter bikes and collapsible bikes to mountain bikes and racing bikes. The advantages of cycling, including its health benefits, low costs and reduced environmental impact, have all increased the appeal of this remarkable invention.

Questions 1 - 9

Complete the notes below.

Choose **ONE WORD ONLY** from the passage for each answer.

Write your answers in boxes 1-9 on your answer sheet.

Early history of the bicycle

The concept of a bicycle

- said to originate with Leonardo da Vinci
- 1 exist whether a drawing of a two-wheeled device was one of da Vinci's

Celerifere

- made of wood and regarded as a 2

Draisienne

- made of wood and could be steered
- required a 3 surface on which it could be ridden

Velocipede

- pedals connected to front wheel
- provided a rough ride outdoors
- 4 were constructed in cities for recreational purposes

High Wheeler/Penny Farthing

- large front wheel provided advantage of greater 5
- tyres were made of 6
- position of 7 made getting on the bicycle difficult
- highly unstable for riding

Ladies' tricycle

- the design meant improved 8 and thus better safety
- allowed women to ride without their 9 causing problems

Questions 10 - 13

Do the following statements agree with the information given in Reading Passage 1? In boxes 10-13 on your answer sheet, write

TRUE if the statement agrees with the information

FALSE if the statement contradicts the information

NOT GIVEN if there is no information on this

10. Leonardo da Vinci built a working model of the bicycle based on his sketches.
11. The velocipede became popular mainly because people enjoyed riding it indoors.
12. The safety bicycle allowed riders to travel at high speeds with less risk of injury.
13. Modern bicycles still retain many design features of John Starley's Rover.



READING PASSAGE 2

You should spend about 20 minutes on Questions 14–26, which are based on Reading Passage 2 below.

Nanotechnology: the science of the very small

A Nanoscience and nanotechnology have become very fashionable in recent times. Their prominence has inspired science fiction and raised a lot of hope and hype about the possibilities. But what exactly is nanotechnology? Most definitions agree that it is the manufacture of structures that measure up to 100 nanometres (nm). The diameter of an atom is typically 0.1 or two or three tenths of a nanometre while that of the human hair is around 1,000,000 nm. Essentially, we are now able to make new structures from atoms and molecules as if we were using the bricks of a children's construction toy.

B Most of the electronic consumer goods that we use today contain components that measure less than 100 nm. The lasers in our CD and DVD players make use of the properties of semiconductor layers that are just a few atoms in thickness, so we clearly are in the nanotechnology age. There are many other everyday items that we take for granted that have nanoparticles in them. One of these is sunscreen. In the 1990s there were safety issues with many of the sunscreen products available, so scientists at Oxford University set about finding a solution. The product they developed uses nanoparticles of titanium dioxide that are doped with the element manganese. This reduces the hazard of potentially skin-damaging agents known as free radicals. The product is almost transparent to the eye when spread on the skin, and it is effective for much longer than many of the traditional preparations. In this case, we see an example of nanotechnology providing a solution to an existing problem rather than opening up new product areas.

C The technique of using nanotechnology to adapt existing products can also be applied to garments, upholstery, paint and plastics, to improve their longevity. Bearing this in mind, it is interesting to ask whether nanotechnology can help with some of the world's major current issues. The recent growth in climate change awareness calls a range of things into question. Can we use energy more effectively? Can we reduce the carbon dioxide in the atmosphere? Can we use and purify water more effectively? Can healthcare be improved? Nanotechnology may very well provide the solutions. There are few times in history when the application of science to improving the human condition has been so apparent.

D We are already making new types of solar cells and batteries by using our newfound ability to manipulate matter at the atomic and molecular scale. If we can capture some of the huge quantities of energy that arrive at the Earth's surface from the Sun, and store it effectively, we can reduce the need for fossil fuel-based power stations. We are already using nanoparticles and nanoporous materials to capture harmful greenhouse gases, and we may possibly convert some

of them back into fuel. There are, however, many technical problems involved, and it will be another decade or so before some of these ideas are deployed.

Water purification is increasingly challenging and many believe that water, which we take for granted, is going to become more valuable than oil. Currently we do not use water efficiently, and many industrial and food processes leave contamination in water that is potentially toxic and difficult to remove. Nanofiltration may provide the solution to these problems, and may also offer new routes to the desalination of brackish water and groundwater in some parts of the world.

E Water purification is increasingly challenging and many believe that water, which we take for granted, is going to become more valuable than oil. Currently we do not use water efficiently, and many industrial and food processes leave contamination in water that is potentially toxic and difficult to remove. Nanofiltration may provide the solution to these problems, and may also offer new routes to the desalination of brackish water and groundwater in some parts of the world.

F Food production and processing are becoming more important as the world's population increases. There are concerns that potentially harmful nano-products might enter the body via the gut. Such concerns are being addressed, but it is even more important to ask questions about how nanotechnology might increase the nutritious yield of food and be adapted to help preserve and package food. There is already evidence to show that it can also be helpful in the primary production of raw food materials, by way of better delivery of pesticides to the field and better extraction of protein and carbohydrates.

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G Healthcare is an area where the application of nanotechnology is already exceeding the expectations of ten years ago. It is happening across several fronts: better appreciation of what the 'natural' nanoparticles circulating in body fluid do; better methods of delivering drugs to the body via nanoparticles that target the diseased body part; improvements to the diagnostic imaging techniques of ultrasound, magnetic resonance imaging and X-ray computerised tomography. We are now at the threshold of a new subject, 'theranostics', whereby nanoparticles will first deliver the diagnostic information to the clinician and then, using the same particle, deliver a form of therapy.

H Nanotechnology is a life-changing technology, making the quality of life better, enabling us to make use of our resources more effectively, thereby protecting our environment, and creating new business opportunities. There are many other examples in the world of 'new materials' that also make use of nanotechnology. The possibilities may well be limitless.

Questions 14–16

Complete the summary. **Write ONE WORD ONLY** from the text for each answer.

Nanotechnology

In the same way that a child may use _____ (14), scientists are now able to create new structures by combining the smallest particles known to man. We can fully appreciate the microscopic scale of this work by comparing the exact _____ (15) of one atom with that of a single human hair.

A lot of today's electronic consumer goods, such as lasers, have components that are microscopic in size. They are not the only products relying on nanotechnology. For example, a long-lasting _____ (16), widely available today, relies on minute nanoparticles for its effectiveness. This was developed in the 1990s as a result of concerns about the safety of existing products.

Questions 17–22

The text has eight paragraphs, A–H.
Which paragraph contains the following information?

- 17. a comparison between two of the world's vital natural resources
- 18. a mention of how nanotechnology could be dangerous for humans
- 19. a claim that nanotechnology has influenced modern authors
- 20. a reference to the fact that nanotechnology has achieved more than was anticipated a decade ago
- 21. a suggestion that nanotechnology could provide financial as well as ecological benefits
- 22. an acknowledgement that more time is required to realise some of nanotechnology's potential

Questions 23–26

Choose **TWO** correct answers. 23–24

Which **TWO** of the following statements does the writer make about nanotechnology and energy resources?

- A. Nanotechnology means that solar energy will replace fossil fuel energy.
- B. We have made solar batteries for the first time by applying nanotechnology.
- C. Further research is needed to turn nanotechnology into practical energy solutions.
- D. Nanotechnology is being used to remove dangerous gases from the atmosphere.
- E. Nanotechnology has helped create cheaper sources of energy.

Questions 25–26

In which **TWO** ways does the writer think nanotechnology could improve food production?

- A. by enabling crops to grow faster
- B. by keeping food fresh for longer
- C. by making food tastier
- D. by protecting food from insects better
- E. by increasing the variety of food

READING PASSAGE 3

*You should spend about 20 minutes on **Questions 27-40**, which are based on Reading Passage 3 below.*

On art and artists

A There really is no such thing as art, only artists. Once these were people who took coloured earth and roughed out the forms of a bison on the wall of a cave; today some buy their paints, and others design posters for hoardings; they did and do many other things. There is no harm in calling all these activities art as long as we keep in mind that such a word may mean very different things in different times and places, and as long as we realize that 'Art' with a capital 'A' has no existence: it has become something of an obsession. You may crush an artist by telling him that what he has just done may be quite good in its own way, only it is not true art.

B Actually, I do not think that there are any wrong reasons for liking a statue or a picture. We may like a painting of a village scene because it reminds us of home. There is nothing wrong with that. All of us, when we see a painting, are bound to be reminded of a hundred-and-one things which influence our likes and dislikes. As long as these memories help us to enjoy what we see, we need not worry.

C It is only when some irrelevant memory makes us prejudiced, when we instinctively turn away from a magnificent picture of an alpine scene because we dislike climbing, that we should search our mind for the reason for the aversion which spoils a pleasure we might otherwise have had. There are wrong reasons for disliking a work of art.

D Most people like to see in pictures what they would also like to see in reality. This is quite a natural preference. We all like beauty in nature, and are grateful to the artists who have preserved it in their works. Nor would these artists themselves have rebuffed us for our taste. When the great Flemish painter Rubens made a drawing of his young son, he was surely proud of his good looks, and wanted us, too, to delight in the child's appearance. But this bias for the pretty and engaging subject is apt to become a stumbling block if it leads us to reject works which represent a less appealing subject.

E The great German painter Dürer certainly drew his mother with as much devotion and love as Rubens felt for his child. His truthful study of old age may give us a shock which makes us turn away from it and yet, if we fight against that first reaction we may be richly rewarded. For Dürer's drawing in its tremendous sincerity is a great work. The beauty of a picture does not really lie in the beauty of its subject matter. The trouble about beauty is that tastes and standards of what is beautiful vary so much.

F What is true of beauty is also true of facial expression. In fact, it is often the expression of a figure in the painting which makes us like or loathe the work. Some people like an expression—perhaps of suffering or of compassion—which they can readily understand, and which therefore moves them profoundly. But we should not, for that reason, turn away from works whose expression is perhaps less easy to understand. Often we must first learn to know the artist's methods of drawing to understand the feelings he or she wishes to portray, and may even come to prefer works of art in which expression is not obvious. Just as some of us would rather people use few words and gestures and leave something to be guessed, so some of us would rather look at paintings or sculptures which are open to interpretation.

G But here newcomers to art are often brought up against another difficulty. They want to admire the artist's skill in representing the things they see. What they like best are paintings which 'look real'. This is an important consideration, and the patience and skill which go into the faithful rendering of the visible world are indeed to be admired. Great artists of the past have devoted much labour to works in which every tiny detail is carefully recorded. Dürer's watercolour study of a hare is one of the most famous examples of this loving patience. But who would say that Rembrandt's drawing of an elephant is necessarily less good because it shows fewer details? Indeed, Rembrandt gave us the feel of the elephant's wrinkly skin with just a few lines of his chalk.

H If we find fault with the accuracy of a picture, we must remember that artists may have their reasons for changing the appearance of what they see. We should also never condemn a work for being incorrectly drawn unless we are quite sure that we are right and the painter is wrong. We have a curious habit of thinking that nature must always look like the pictures we are accustomed to. For generations, paintings showed horses galloping with outstretched legs in full flight through the air, and we came to believe that this was indeed how they moved. It was only the development of photography which proved that horses simply do not gallop in that way. And yet, when painters began to apply this new discovery, and painted horses moving as they actually do, everyone complained that their pictures looked wrong.

Questions 27-30

Which paragraph contains the following information?

*Write the correct letter, **A-H**, in boxes 27-30 on your answer sheet.*

- 27.** A comparison between people's preferences towards language and art.
- 28.** Why people may have misunderstood how a certain animal moves.
- 29.** A contrast between the techniques used in two pictures.
- 30.** A possible negative effect of using the term 'art'.

Questions 31-36

*Match each statement with the correct subject, **A-G**. Write the correct letter in boxes 31-36 on your answer sheet.*

- 31.** The artist clearly hoped that we would share his admiration of this subject.
- 32.** The artist aims to give a general impression of his subject rather than a detailed one.
- 33.** Someone's personal experience may make them consider a painting of this subject unpleasant.
- 34.** The way in which this subject was once portrayed was factually wrong.
- 35.** The painting is well known for the effort the artist put into creating a very accurate picture of this subject.
- 36.** People may like a painting of this subject for a reason that is external to the painting.

List of Subjects

- A.** A village scene
- B.** Alpine scene
- C.** Rubens's son
- D.** Dürer's mother
- E.** Hare
- F.** Elephant
- G.** Horses

Questions 37-40

Complete the summary below. Choose **ONE WORD ONLY** from the passage for each answer.

Beauty and expression

Differences in tastes and standards of beauty make it difficult to judge a painting. Dürer's painting of his mother is considered significant because it is a **37** representation of her age and situation. However, we may need to overcome an initial reaction of **38** when we see it.

In order to judge the facial expressions of people portrayed in a painting, we need to become familiar with the methods that the artist has used when creating the painting. While some of us are affected by the portrayal of a facial expression that we can **39** without much effort, others prefer expressions that require **40**

